COMPLETE IMPLEMENTATION AUDIT & AGENT DIRECTIVE

Project: DocSummarize — AI Document Summarization System
Department: Department of Computer Science, Gombe State University
Target: Final Year Undergraduate Project Report (Chapters 1 to 5)

SECTION 1: CODEBASE INVENTORY & FILE-BY-FILE AUDIT

The project repository (`AI-Summary`) is clean, tested, and contains **no dead files, no fake features, and no ungrounded dependencies**:

File / Folder	Lines	Exact Implemented Responsibility
`app.py`	514	**Flask Web Controller & Job Orchestrator:** Handles file uploads, sliding-window IP rate limiting (10 req/min), asynchronous worker dispatch (`threading.Thread`), live polling endpoint (`/status/<job_id>`), thread cancellation (`/cancel/<job_id>`), image streaming (`/image/<job_id>/<index>`), and export streaming (`/download/<job_id>`). Cleans up completed jobs after 1 hour (`JOB_CLEANUP_AFTER_SECONDS = 3600`).
`config.py`	66	**Central Configuration:** Sets 20 MB file size limit (`MAX_CONTENT_LENGTH`), 900-token chunk budget, 60-token overlap, and presets: Short (25%), Medium (40%), Long (60%). Sets adult reading speed (`READING_WPM = 238`).
`run.py`	56	**Production Entry Point:** Single-command script that launches the Flask application on `http://127.0.0.1:7860` and automatically opens the user's default browser.
`requirements.txt`	8	**Dependencies:** `flask`, `werkzeug`, `transformers` (used for tokenizer only), `PyMuPDF` (fitz), `python-docx`, `reportlab`, `gunicorn`. *Note: Heavy neural libraries like `torch` were audited and removed because no neural generation is performed.*
`utils/pdf_extractor.py`	730	**PDF Parsing & Multi-Modal Engine:** Extracts digital text blocks via PyMuPDF (`fitz`). Identifies raster images (≥60px×60px≥60px×60px) and coordinates for diagram anchoring. Detects monospaced font descriptors (`Courier`, `Consolas`, `Mono`) to isolate code blocks verbatim. Skips publisher Table of Contents (TOC) front-matter pages (pages 1–25).

File / Folder	Lines	Exact Implemented Responsibility
`utils/docx_extractor.py`	98	**Word Document Ingestion:** Extracts paragraph text and table cell text from `.docx` files using `python-docx`.
`utils/txt_extractor.py`	18	**Plain Text Ingestion:** Reads raw UTF-8 text files.
`utils/text_cleaner.py`	842	**Sanitization & Metadata Engine:** Extracts metadata (Course Code, Title, Lecturer, Date) from the first 40 lines. Removes repetitive slide headers (occurring on $\geq 3 \geq 3$ pages). Executes the 5-layer ZapfDingbats (■) black-box fix. Normalizes bullets to clean markdown dashes (`-`).
`utils/chunking.py`	200	**Sentence Splitter & Token Chunker:** Abbreviation-aware sentence splitter (protects `Dr.`, `Prof.`, `e.g.`, decimals). Chunks text into $\leq 900 \leq 900$-token windows using the BART tokenizer with a 60-token sliding overlap.
`utils/summarizer.py`	1,414	**Core Extractive Engine:** Computes sub-linear TF-IDF vectors, sentence centrality, definition bonuses (+0.35), takeaway bonuses (+0.30), and Maximal Marginal Relevance (MMR, $\lambda = 0.65 \lambda = 0.65$). Identifies section headings (`##`, `###`, `####`), pairs balanced comparative differences, and extracts the top 5 key highlights.
`utils/keywords.py`	148	**Keyword Engine:** Extracts up to 8 domain keywords using sub-linear TF-IDF weighting without external API calls.
`utils/file_validator.py`	45	**Security Validator:** Enforces extension validation (`pdf`, `docx`, `txt`), MIME-type validation, and the 20 MB size ceiling.
`utils/download_generator.py`	799	**Multi-Format Report Generator:** Builds downloadable files on demand: ReportLab PDF (with custom two-pass `PageNumCanvas` for "Page X of Y", embedded diagrams, and headers), Word DOCX (with native styles), and Plain Text.
`utils/helpers.py`	12	**Shared Utilities:** UUID4 job ID generator and word count counter.
`tests/`	7 files	**Automated Test Suite:** 17 unit and integration tests covering chunking, docx, pdf, download generation, routes, summarization, and txt ingestion. **All 17 tests pass 100% in 12.15 seconds.**

SECTION 2: THE 5 REAL CONTRIBUTIONS OF THE PROJECT

When writing or reviewing the report, the agent must frame the student's work around these **five specific engineering contributions**:

1. **Automatic Diagram & Monospaced Code Preservation:**
    - In technical education, textbooks and lecture notes depend on diagrams and code.

- While existing tools strip images and break programming syntax, this software isolates monospaced code blocks so they bypass NLP summarization (retaining 100% exact indentation), and extracts raster diagrams ($\geq 60\text{px} \geq 60\text{px}$) to anchor them directly under their corresponding summary topics.

2. **Hand-Crafted Extractive Ranking Engine (Zero Hallucination):**
   - Built in pure Python without generic third-party summarization wrappers.
   - Ranks sentences using a multi-factor formula that boosts definitions (`is defined as`, `refers to`) and core takeaways (`primary objective`, `key concept`), filtered with Maximal Marginal Relevance (MMR, $\lambda = 0.65 \lambda = 0.65$) to eliminate duplicate points.
3. **The 5-Layer Table of Contents (TOC) & Black-Box (■) Fix:**
   - Diagnosed that publisher dot leaders in textbooks use `ZapfDingbats` fonts that render as solid black squares (■) and pollute summaries.
   - Engineered a 5-layer defense across PDF extraction, text cleaning, summarizer scoring, ReportLab canvas sanitization, and frontend UI rendering to permanently eliminate the issue.
4. **Balanced Comparative Difference Extraction:**
   - When summarizing sections discussing differences (e.g. *Differences between Tightly Coupled and Loosely Coupled Systems*), generic tools pick points from one side only.
   - Engineered an algorithm (`_extract_comparison_points`) that identifies Subject A and Subject B and pairs contrasting points into balanced statements:
     "*Subject A: [Point A], whereas Subject B: [Point B].*"
5. **Asynchronous Architecture with Multi-Format Export:**
   - Implemented background worker threads with dynamic 1.2s status polling and safe thread cancellation.
   - Designed a two-pass ReportLab canvas (`PageNumCanvas`) for dynamic "Page X of Y" numbering with custom margins and embedded figures.

SECTION 3: MINIMAL & CLEAN CALCULATIONS (KEEP MATH BRIEF)

The user requested that calculations **do not appear excessively** in the report. Include **only these three concise equations**:

1. **Sentence Saliency Scoring Equation (Chapter 3):** $\text{BaseScore}(Si) = (0.30 \cdot \text{Centrality}i + 0.25 \cdot \text{Specificity}i + 0.15 \cdot \text{Density}i + 0.30 \cdot \text{Bonus}i) \times \text{PositionWeight}i$ $\text{BaseScore}(S_i) = (0.30 \cdot \text{Centrality}_i + 0.25 \cdot \text{Specificity}_i + 0.15 \cdot \text{Density}_i + 0.30 \cdot \text{Bonus}_i) \times \text{PositionWeight}_i$ *(Explains how definitions and key concepts are prioritized over filler words).*
2. **Diversity & Non-Redundancy via MMR (Chapter 3):** $\text{MMR}(Si) = 0.65 \cdot \text{BaseScore}(Si) - (1 - 0.65) \cdot \max_{Sj \in \text{Selected}} \text{Sim}(Si, Sj)$ $\text{MMR}(S_i) = 0.65 \cdot \text{BaseScore}(S_i) - (1 - 0.65) \cdot \max_{S_j \in \text{Selected}} \text{Sim}(S_i, S_j)$ *(Explains how duplicate sentences covering the same point are suppressed).*
3. **Summary Word Reduction Percentage (Chapter 4):** $\text{Reduction Percentage} = \left(1 - \frac{\text{Summary Words}}{\text{Original Words}}\right) \times 100\%$ *(Tested on `COSC 414 M1-2026.pdf`: reduced from 1,220 words to 531 words = **56.5% reduction** in **1.4 seconds**).*

STRICT RULE: Do NOT include ungrounded ROUGE-1/ROUGE-2 decimal tables, neural network loss calculations, or gradient descent math, as the system uses an extractive pipeline rather than training a neural model.

SECTION 4: SYSTEM LIMITATIONS (FOR CHAPTER 1 & 5)

Document these real, authentic technical boundaries:

1. **No Built-in OCR:** The software extracts digital text streams. Scanned image-only PDFs without an OCR text layer cannot be parsed directly and require external OCR pre-processing.
2. **Diagram and Code Extraction Bound to PDF:** Visual diagrams and monospaced code blocks are extracted from PDF files only; DOCX and TXT files extract text only.
3. **No Complex Table Reconstruction:** Embedded spreadsheets and multi-column tables are extracted as linear text lines rather than reconstructed into Markdown/HTML table matrices.
4. **Verbatim Extractive Boundary:** The system does not paraphrase, reword, or repair poor author grammar.
5. **Monolingual Optimization:** Stopwords and definition regex patterns are optimized for English.
6. **In-Memory Volatile Job Storage:** Jobs reside in server RAM with a 1-hour Time-To-Live (TTL); multi-server horizontal scaling requires adding Redis.
7. **20 MB Upload Ceiling:** Files larger than 20 MB are rejected to protect local server memory.

SECTION 5: RECOMMENDATIONS FOR FUTURE WORK (FOR CHAPTER 5)

1. **Embedded OCR Pipeline:** Integrate Tesseract OCR (`pytesseract`) directly into `pdf_extractor.py` for scanned image PDFs.
2. **Distributed Queue Backend:** Migrate the in-memory job store (`_jobs`) to Redis and Celery for enterprise multi-server load balancing.
3. **Table Lattice Extraction:** Incorporate `pdfplumber` or `Camelot` to recognize table borders and output formatted Markdown tables.
4. **Word Document Drawing Parser:** Extend `docx_extractor.py` using `docx.oxml` to extract inline drawings and shapes from Word files.
5. **Multilingual Stopword Expansion:** Integrate NLTK multi-language corpora to support technical documents in other languages.
6. **Persistent User Accounts:** Introduce user authentication and a relational database (PostgreSQL/SQLite) for multi-device summary history.

SECTION 6: DIRECTIVE FOR YOUR WRITING AGENT

Give your writing agent this exact instruction:

text

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Please review the existing final year project report draft (or write Chapters 1 through 5)
based on the "COMPLETE IMPLEMENTATION AUDIT & AGENT DIRECTIVE" above.
Enforce these strict standards:
1. Ensure the report reflects the actual implementation: Pure Extractive Summarization using
Sub-linear TF-IDF + MMR (λ = 0.65) with Diagram and Code Block Preservation.
2. Remove any claims that the system uses neural generative models like BART to generate text,
or that it summarizes tables.
3. Highlight the student's 5 core contributions in Chapters 1, 3, 4, and 5.
4. Keep calculations minimal and clean: include only the 3 essential formulas provided
(Saliency Scoring, MMR, and Reduction Percentage). Do not add fake ROUGE tables or complex
neural loss math.
5. Accurately document the system limitations and future recommendations in Chapters 1 and 5.
6. Cite the real case study in Chapter 4: COSC 414 Module 1 (1,220 words reduced to 531 words,
56.5% reduction, 1.4s CPU runtime, Figure 1.1 and 1.2 preserved, 17/17 passed automated
tests).
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